

Element C	Informaiton	Other Details	Date Added
C1	- Needs to be fit to hold a piece of paper (8.5x11 inches). The blood splatter lab involves dripping the blood onto a piece of standard paper, which is why the apparatus must fit and secure the standard piece of paper. - Needs to be lightweight and easily transportable by students (under 5 lbs). The device will be used by multiple groups and will be shared between tables, meaning it must be easy to transport. - Must also be sized smaller than the science lab table itself (30x48 inches), but preferably much smaller - Must have a secure paper-holding mechanism (no slipping - therefore invalidating the blood pattern results). If the paper were to slip or move, the lab's result would be ruined. - Contains the ability to increment the angle by 5-10 degrees with clear angle indicators on the apparatus. The ability to drop the blood at specific increments is needed to give the students the best and most valuable lab experience. - Should last/not be easily destroyed by students during use (durable and long-lasting 1+ years). Students in the class may drop or mishandle it, it needs to be able to withstand the abuse. - Able to be constructed for less than \$50, our group will be given a specific budget for materials, and the device needs to be able to be built within the budget. - Contains the function to easily attach and detach the paper from the apparatus - easy to reset between trials (under 30 seconds). For groups to have the best lab experience, they will need to have the ability to run multiple tests from different angles, therefore, the paper will need to be removed easily. - Visually appealing/having a "sleek look", ALSO containing "educational/clear labeling (for classroom clarity, helping students understand its parts and purpose) - Stability during use (any wobble will distort droplet data). Distorted data would lead to a failure to recreate the crime scene, which would harm students' education. - It must be Safe for users of varying skill ranges (no sharp edges or pinch points). Sharp points or pinch points could lead to minor injuries and cuts, which would disrupt students' education. - No electronic aspects are required (keeps it affordable and low-maintenance)		4/4/25
C2	- Angle adjustment from 0° to 90° in 10° increments (Core function of the apparatus) - Stability during use (otherwise, any wobble may distort droplet		4/4/25

	data) - Must contain a secure paper-holding mechanism (slipping also invalidates blood pattern results). - Must cost less than \$50 per unit (keeps the project scalable and available to all schools). - Easy to reset between trials (under 30 seconds) - speeds up classroom workflow. - Durable and long-lasting (1+ years of use) - must last through repeated classroom use/multiple classes using it throughout the lab each year. - Lightweight (under 5 lbs) - Easy for students to move around. - Safe (no sharp edges or pinch points) - student safety in the school lab is certainly a priority, although a project such as this sees little risk of injury, etc. - Can securely grasp any paper input into the device (different weights/heights/thicknesses of paper may be used). - No electronics are required (It is unlikely that a project like this will need electronic aspects anyway). - Aesthetic/education labeling (for classroom clarity) - helps students understand the parts and their purpose during the lab. - Blood must not soak into the surface of the material from which the apparatus is made (non-porous).		
C3	<u>High Importance</u> - Angle adjustment from 0° to 90° in 5° increments (Core function of the apparatus), [measuring] the accuracy of angle using a protractor (consult: Forensic Science Teacher) A - Must contain a secure paper-holding mechanism (slipping also invalidates blood pattern results). [measuring] the dimensions of the table it sits on/paper it holds with a ruler or measuring tape (consult: Forensic Teacher) B - Must cost less than \$50 per unit/imposed budget on E4USA Class (keeps the project scalable and available to all schools). [measuring] the cost of materials/every item required for its construction (consult: Engineering Teacher) C - Easy to reset between trials (under 30 seconds) - speeds up classroom workflow. [measuring] the time to reset each with a stopwatch (consult: Forensic Teacher/student “test group”) D - Durable and long-lasting (1+ years of use) - must last through repeated classroom use/multiple classes using it throughout the lab each year. Perhaps stimulated through a “wear-and-tear” test (consult: Forensic Science teacher/Lab Tech/Students using the apparatus) E - Safe (no sharp edges or pinch points) - student safety in the school lab is certainly a priority, although a project such as this sees little risk of injury, etc. Test Sharp edges visually/by touch (consult: Teacher/Students participating in the lab) F <u>Low Importance</u> - Lightweight (under 5 lbs) - Easy for students to move around. [measuring] the total weight by using a scale (consult:		4/10/25

	<u>Students/teachers</u>) G - Stability during use (otherwise, any wobble may distort droplet data), [measuring] the Force needed to tilt/disrupt the apparatus (<u>consult: Engineering teacher or Physics teacher</u>) H - No electronics are required (It is unlikely that a project like this will need electronic aspects anyway). Confirm it functions manually/no automation necessary (<u>consult: Forensic Teacher/Engineering Teacher</u>) I - Aesthetic/education labeling (for classroom clarity) - helps students understand the parts and their purpose during the lab. [analyze] The labels and their clarity/usefulness (<u>consult: Forensic Science Teacher & Students participating in the lab</u>) J - Blood must not soak into the surface of the material from which the apparatus is made (non-porous). Material surface testing - wiping the surface clean (<u>consult: Forensic Teacher/Engineering Teacher</u>) K - If time permits, multiple iterations may be produced to fit the needs of multiple students during the completion of this lab. L		
C4	- Angle adjustment from 0° to 90° in 5° increments can be measured using a protractor (the accuracy of angle) (<u>consult: Forensic Science Teacher</u>) - We may measure the stability during use with the Force needed to tilt/disrupt the apparatus (<u>consult: Engineering teacher or Physics teacher</u>) - The secure paper-holding mechanism - the dimensions of the table it sits on/the paper it holds, may be measured with a ruler or measuring tape (<u>consult: Forensic Teacher</u>) - Must cost less than \$50 per unit/imposed budget on E4USA Class. We can quantify the cost of materials/every item required for its construction. (<u>consult: Engineering Teacher</u>) - Easy to reset between trials (under 30 seconds) - speeds up classroom workflow. Can be measured by the time it takes to reset each paper with a stopwatch (<u>consult: Forensic Teacher/student "test group"</u>) - Durable and long-lasting (1+ years of use) - must last through repeated classroom use/multiple classes using it throughout the lab each year. Measured/Analyzed the effects through a stimulated "wear-and-tear" test or analyzed the apparatus after multiple uses. (<u>consult: Forensic Science teacher/Lab Tech/Students using the apparatus</u>) - Lightweight (under 5 lbs) - Easy for students to move around. We can measure the total weight of our design by using a scale (<u>consult: Students/teachers</u>) - Safe (no sharp edges or pinch points) - student safety in the school lab is certainly a priority, although a project such as this sees little risk of injury, etc. We can analyze/test the sharp edges through visual analysis/by physical touch (<u>consult: Teachers/Students participating in the lab</u>) - No electronics are required (It is unlikely that a project like this will need electronic aspects anyway). Confirm it functions		4/7/25

	manually/no automation necessary (<u>consult: Forensic Teacher/Engineering Teacher</u>) - Aesthetic/education labeling (for classroom clarity) - helps students understand the parts and their purpose during the lab. We may analyze the labels and their clarity/usefulness as other students try to interpret their meaning. (<u>consult: Forensic Teacher & Students participating in the lab</u>)		
C5	● Angle Adjustment Accuracy - The apparatus must allow paper to be positioned at angle increments of exactly 5 degrees, ranging from 0° to 90°, to enable accurate data collection. ● Secure Paper Holding Mechanism - The apparatus must securely hold standard 8.5" x 11" paper without slipping when angled, to ensure consistency in spatter patterns. ● Angle Locking Feature - Each angle setting must have a locking mechanism that prevents unintentional movement once the angle is selected. ● Durability Under Use - The materials used must withstand daily classroom use for at least one semester (approx. 1 year) without structural failure or degradation. ● Easy Adjustment by Students - The angle of the apparatus must be adjustable in under 30 seconds by a high school student without tools, ensuring usability during class time. ● Clear Angle Markings - The apparatus must have visible, labeled markings at each 5° increment to allow precise and fast adjustments during experiments. ● Stability on Flat Surfaces - The base of the apparatus must remain stable and unmoved on flat lab tables during use to prevent data inconsistencies. ● Fits Within Desk Space - The entire apparatus must fit within a table space, ensuring compatibility with typical classroom lab tables. ● Cost-Effective for Classroom Use - The estimated cost per unit must not exceed \$50 to ensure affordability for multiple classrooms to implement.		4/7/25
C6	Dr. Kovacs approved of all requirements/criteria, mentioning that some details that we had included were beyond what he initially thought as well. Dr. Kovacs exclaimed, "Oh, I didn't think of that!" in reference to our listed criteria and basic ideas for the upcoming project. Based on the requirements provided, new ideas were therefore inspired as we		4/10/25

	prepare to move on to the next stages of the engineering design. We questioned whether this product may expand the lab and provide more opportunities in further trials for years to come, and he responded with an astounding "Yes!" With more accurate data collection and replicable results, the lab will be more accessible to various Forensic Science classes around MCPS and will result in a more in-depth focus on this Blood Splatter Lab -Dr. Kovacs. He ultimately concluded that our insight was helpful in expanding upon our initial intentions, although we should focus specifically on holding an 8.5" x 11" piece of paper with adjustable angles throughout, any additions being ancillary.		
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